Demonstrations are weights you write at inference time

And the non-negativity that makes those weights readable is what makes them forget.

DATAFORGE 2026, PATHWAY TRACK — ONE-PAGE CONCEPT SUMMARY. TOPIC: TEST-TIME ADAPTATION, OPTIMIZATION VERSUS CONTEXT.

The design pressure

A model meeting an unfamiliar task has two routes. It can **optimize** — turn the demonstrations into training samples and take a backward pass first, as HRM (arXiv:2506.21734) and TRM (arXiv:2510.04871) do on ARC at ~$1.48–1.76 per task. Or it can

adapt in state: write them into a recurrent memory and answer with every parameter untouched. BDH-CQ takes the second route three orders of magnitude cheaper. The question is not which is cheaper, but what the second route *charges* instead.

The mechanism, and the part everyone guesses wrong

A softmax-free attention with `Q = K` has an exact recurrent form. Where a Transformer keeps a KV cache that grows with every token, the same computation can be carried in a fixed-size associative state updated by rank-one writes:

$$\sigma_t = \sigma_{t-1} + \mathrm{rope}(K)_t \otimes V_t$$
$$\mathrm{out}_t = \sigma_t^{T}\, \mathrm{rope}(Q)_t$$

Not an approximation: the official MIT `bdh.py` run both ways agrees to **2.8×10^{-14} in float64**, and our 8M model's recurrent decode reproduces its parallel forward to **6×10^{-7}**. The natural assumption is that BDH's three unusual choices — no softmax, `Q = K`, a ReLU forcing non-negative activations — all buy that constant memory.

They do not. Softmax breaks the equivalence (`1.2 × 10²`); `Q ≠ K` (`1.7 × 10⁻⁵`) and dropping the ReLU (`1.6 × 10⁻⁵`) leave it intact. Softmax-freeness buys the fixed state. `Q = K` and the ReLU buy *readability*.

What readability costs

Non-negative activations make BDH inspectable — sparse, positive, monosemantic structure is the Dragon Hatchling paper's claim (arXiv:2509.26507). On an 8M BDH we trained, 4.93% of units clear 1% of their peak against GPT-2's 86.3%, and `G* = Dₓᵀ Eᵀ` predicts

from the weights alone, with no data, which neurons stay silent on real text: MCC **+0.944**, against **−0.001** for a shuffled null and **+0.908** on an independent Portuguese sibling.

That same non-negativity caps what the state can hold. Writing *k* rank-one bindings into a fixed N×D state and reading them back gives 100% at k=8, 94% at k=32, 47% at k=128. The counterfactual identifies the cause: with **signed** keys the identical state holds 384 bindings at 99%, past its own rank. Mean pairwise cosine is **+0.319 for ReLU'd keys versus −0.000 for signed**, and +0.319 is analytic — **$1/\pi$** for rectified Gaussians. Non-negative vectors cannot be near-orthogonal, so they interfere. Interpretability is paid for in memory, at a fixed rate. The ceiling belongs to the linear-attention family, not to BDH: DeltaNet (arXiv:2406.06484) replaces "the additive update in linear transformers" — BDH's write — with a delta rule, reporting it better at associative recall. Ours is the mechanism and the constant.

	KV-cache Transformer	BDH (public, MIT)	BDH-CQ	HRM / TRM
Memory per extra token	grows linearly	**constant**, cheaper past `N_total/2`	constant (dims undisclosed)	no context state
Adaptation lives in	the context window	recurrent state σ	recurrent state	**weights** — a backward pass
Inspectability	attention maps	**4.93%** active + a synapse graph	none published	none
Evidence here	cost only	**we train and run it**	replayed, cited	authors' report

GPT-2 is not a control. The contrast that is structural rather than empirical: **38.7%** of BDH's neurons are isolated in `G*` and they are the silent ones, against **0.1%** in GPT-2's nearest analogue, which maps *two different* neuron populations. Weight-tying is what makes the question well posed.

The roles of BDH and BDH-CQ

BDH is the public substrate — MIT-licensed, runnable, and what every number above was measured on. **BDH-CQ is evidence only**: no weights, no API, dimensions and update rules "remain proprietary." The bridge is exact: BDH-CQ's update is Eq. (1) `Sₜ = U_θ(Sₜ₋₁, Dₜ)`, and §3.2 names its special case `Sₜ = Sₜ₋₁ + U_θ(Dₜ)` as "linear attention being the conceptually simplest standalone realization." **That special case is exactly what `bdh.py` computes.** One correction: additive accumulation is the *named special case*, not BDH-CQ's rule — calling it BDH-CQ's mechanism, as secondary summaries do, is an error.

Reading the evidence honestly

BDH-CQ's headline is 29.5% pass@2 (118/400) on public ARC-AGI-1 at ~0.85 H200 GPU-s/task (arXiv:2608.09888, Table 1). Four qualifications, all from the report:

- **$0.00070/task is computed, not billed** — 0.85 s × an assumed $3/H200-hour. §6.6 reports the *same* 118/400 at **$0.00265246**, 3.8× higher, unreconciled. There, "57× cheaper" is ~15×.
- **Not equal-accuracy**: GPT-5.6 Luna (Low) scores *higher* at 34.2%, and the Wilson interval on 29.5%, [25.24, 34.15], overlaps it.
- The audit reproducing 29.5% was by **co-authors** — developer-reported, not external. BDH-CQ appears on no ARC Prize leaderboard.
- **ConceptARC is in the training mixture** (§4.2) and then evaluated on (59.38%); the paper says its control does not rule out training exposure.

The sharpest limitation is structural, not numerical. Latent reasoning returns grids, not traces, so — the report's words — "a correct rule applied incompletely is observationally indistinguishable from a narrower rule applied completely." Table 9: 89.7% of failures still have correct output dimensions. Not verbalizing is what makes it cheap, and why you cannot audit it.

Continue at https://demonstrations-are-weights.vercel.app, then the Dragon Hatchling paper, BDH-CQ §3.2 and Appendix A.3–A.4, and DeltaNet.

Provenance. Every figure attributed to us is reproducible from **github.com/snivellus38/DataForge** by a named script; every BDH-CQ figure carries a section or table locator into arXiv:2608.09888 and is never recomputed here. `vendor/bdh.py` is Pathway's, MIT and unmodified. Our models are a 131K BDH trained for this submission and an 8M BDH trained by one author beforehand and reused with disclosure (it adds a learned positional embedding over RoPE and has no decay term). Neither is an official BDH model; neither is BDH-CQ. AI-assisted, disclosed in the README.